

OBSERVATION OF 30% CONTINUOUS-WAVE TWO-PHOTON OPTICAL AMPLIFICATION

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Two-photon lasers have intrigued researchers for years because the dynamical behavior of this new class of highly nonlinear oscillator is very different from normal one-photon lasers. Experimental tests of the numerous, often conflicting, predictions regarding their behavior has been hindered by the lack of suitable gain media until the recent development of the dressed-atom two-photon gain medium. We observe $\sim 30\%$ two-photon optical amplification in a new type of laser-driven gain medium that is based on two-photon Raman scattering. This is ~ 300 times larger than previously reported dressed-atom gain.¹ The enhancement is due to the relative insensitivity of the scattering process to broadening mechanisms allowing us to use a high-density atomic vapor.

Large two-photon amplification is observed in a laser-pumped potassium vapor (length 7.5 cm, vapor pressure $\sim 5.5 \times 10^{-4}$ torr). The pump laser (frequency ω_P), an 850 mW actively stabilized Ti:sapphire laser, is linearly polarized, has a ~ 1 mm spot size as it passes through the vapor cell, and is tuned ~ 800 MHz below the 770-nm $4S_{1/2} (F=2) \rightarrow 4P_{1/2} (F=1)$ ^{39}K transition. The pump laser serves two purposes: it creates the necessary inversion between the ground-state hyperfine levels (splitting $\Delta_{hfs} = 462$ MHz) by optically pumping atoms from the $F=2 \rightarrow F=1$ levels as they move into the pump beam; and it acts as the pump field for the two-photon Raman process. The probe laser (frequency ω_R), a grating stabilized diode laser, is polarized orthogonal to the pump laser, crosses the pump laser at a ~ 12 mrad angle, and has a ~ 500 μm beam spot size. Significantly smaller crossing angles (< 6 mrad) could not be used due to the appearance of four-wave mixing processes. We observe large ($> 5,000\%$) amplification of the probe laser due to one-photon Raman scattering² when $\omega_R \approx \omega_P - \Delta_{hfs}$ and its power is low (< 10 μW). For higher powers (~ 8 mW), the one-photon Raman gain decreases dramatically and a new gain feature appears when $\omega_R \approx \omega_P - \Delta_{hfs}/2$, as seen in the figure. We attribute this probe dependent feature to the two-photon Raman scattering process.³ The probe beam is amplified by $\sim 30\%$

in a single pass through the cell which should be sufficient to observe two-photon lasing. Note also the feature at $\omega_R \approx \omega_P - \Delta_{hfs}/3$ which we attribute to three-photon Raman amplification ($\sim 5\%$ gain).

Our current efforts are directed toward suppressing the minor competing effects that occur simultaneously in the system: mild self-focusing and the background caused by the one-photon dressed-atom gain. The efficiency of these processes relative to the Raman process can be reduced by collisionally dephasing the $4S_{1/2} \rightarrow 4P_{1/2}$ transition using a buffer gas since the dephasing rate of the electronic transition is large in comparison to the dephasing rate of the coherence between the ground-levels. Once we have optimized the Raman process, we will investigate how the photon noise of a beam of light is modified by the two-photon gain process and we will develop a two-photon laser based on the Raman process.

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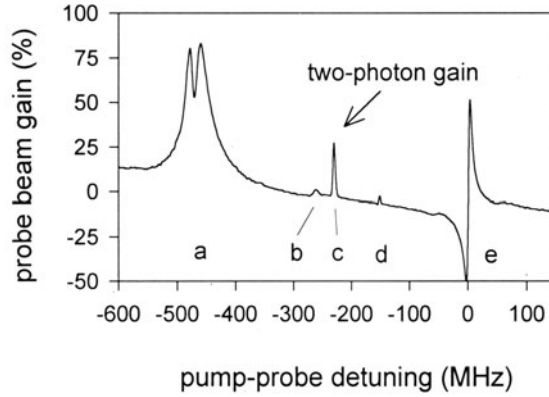


Figure 1 Amplification experienced by a strong probe beam passing through a laser pumped potassium vapor. We identify the spectral features as: a) saturated one-photon Raman gain² from ^{39}K at $\omega_R \approx \omega_P - \Delta_{hfs}$; b) saturated one-photon Raman gain from ^{41}K (6% abundance, 254 MHz hyperfine splitting); c) two-photon Raman gain³ from ^{39}K at $\omega_R \approx \omega_P - \Delta_{hfs}/2$; d) three-photon Raman gain from ^{39}K at $\omega_R \approx \omega_P - \Delta_{hfs}/3$; and e) stimulated Rayleigh gain⁴ near $\omega_R \approx \omega_P$ due to population oscillations between the hyperfine levels. The small, spectrally broad amplification is due to the one-photon dressed-state gain process.

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